



Sir Syed University
of Engineering & Technology

Computer Network Lab

Project Report

Supervised by

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Computer Network Lab Project

BS Computer Science, Batch 2022F

Company Network Design & Implementation Using Cisco Packet Tracer

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Project Report

Introduction

In today's fast-paced business environment, a well-designed network infrastructure is crucial for the seamless operation of any organization. This project focuses on designing a robust, scalable, and secure network for a trading floor support center that is expanding to a new building. The new building will accommodate 600 staff members across three floors, each housing two departments. The network design must meet the current business needs while being future-proofed to accommodate potential growth and changes.

Project Overview

The project involves designing a logical network architecture for a new building with three floors, each housing two departments. The network must support 600 users and 12 server room devices, with specific requirements for each department, including wireless networking, VLAN segmentation, and dynamic IP addressing. The network design must also incorporate redundancy, security, and efficient routing mechanisms to ensure high availability and performance.

Key Components:

- **Three Floors:**
 - **First Floor:** Sales & Marketing (120 users), Human Resource & Logistics (120 users)
 - **Second Floor:** Finance & Accounts (120 users), Administration & Public Relations (120 users)
 - **Third Floor:** ICT (120 users), Server Room (12 devices)
- **Network Requirements:**
 - Hierarchical network design with redundancy at every layer.
 - Wireless networking for each department.
 - VLAN segmentation and subnetting for each department.
 - Dynamic IP addressing via DHCP for all devices except server room devices.
 - Inter-VLAN routing using multilayer switches.
 - OSPF as the routing protocol.
 - SSH for remote access to routers and switches.
 - Port security for the Finance & Accounts department.
 - PAT with ACLs for internet access.
 - Connectivity to two ISPs for redundancy.

Project Objective

The primary objective of this project is to design and implement a logical network architecture that meets the current and future needs of the organization. The network design should ensure high availability, security, scalability, and efficient routing, while also providing a seamless user experience across all departments.

Features

1. Hierarchical Network Design

The network is designed using a hierarchical model, which includes:

- **Core Layer:** High-performance routers (R1 and R2) providing redundancy and connectivity to ISPs.
- **Distribution Layer:** Multilayer switches (MLS-W1 and MLS-W2) providing inter-VLAN routing and redundancy.
- **Access Layer:** Access switches providing connectivity to end-user devices.

2. Redundancy

- **Router Redundancy:** Two routers (R1 and R2) are used to provide redundancy at the core layer.
- **Switch Redundancy:** Two multilayer switches (MLS-W1 and MLS-W2) are used to provide redundancy at the distribution layer.
- **ISP Redundancy:** Connectivity to two ISPs to ensure internet redundancy.

3. VLAN Segmentation

- Each department is assigned to a different VLAN:
 - **VLAN 10:** Sales & Marketing
 - **VLAN 20:** Human Resource & Logistics
 - **VLAN 30:** Finance & Accounts
 - **VLAN 40:** Administration & Public Relations
 - **VLAN 50:** ICT
 - **VLAN 60:** Server Room

4. Subnetting

- The base network 172.16.1.0 is subnetted to provide sufficient IP addresses for each department:
 - **First Floor:**
 - **Sales & Marketing:** 172.16.1.0/25
 - **Human Resource & Logistics:** 172.16.1.128/25
 - **Second Floor:**
 - **Finance & Accounts:** 172.16.2.0/25
 - **Administration & Public Relations:** 172.16.2.128/25
 - **Third Floor:**
 - **ICT:** 172.16.3.0/25
 - **Server Room:** 172.16.3.128/28

5. Dynamic IP Addressing

- DHCP servers located in the server room will dynamically assign IP addresses to all devices except those in the server room, which will have static IP addresses.

6. Inter-VLAN Routing

- Multilayer switches (MLS-W1 and MLS-W2) are configured for inter-VLAN routing, allowing devices in different VLANs to communicate with each other.

7. Routing Protocol

- OSPF (Open Shortest Path First) is used as the routing protocol to advertise routes between routers and multilayer switches.

8. SSH for Remote Access

- SSH (Secure Shell) is configured on all routers and multilayer switches to provide secure remote access.

9. Port Security

- Port security is configured for the Finance & Accounts department to allow only one device to connect to a switch port, using a sticky method to obtain MAC addresses and shutting down the port in case of violation.

10. PAT with ACLs

- PAT (Port Address Translation) is configured to use the outbound router interface IPv4 address, with necessary ACL rules to control internet access.

11. Wireless Networking

- Each department has a wireless network configured to provide wireless connectivity to users.

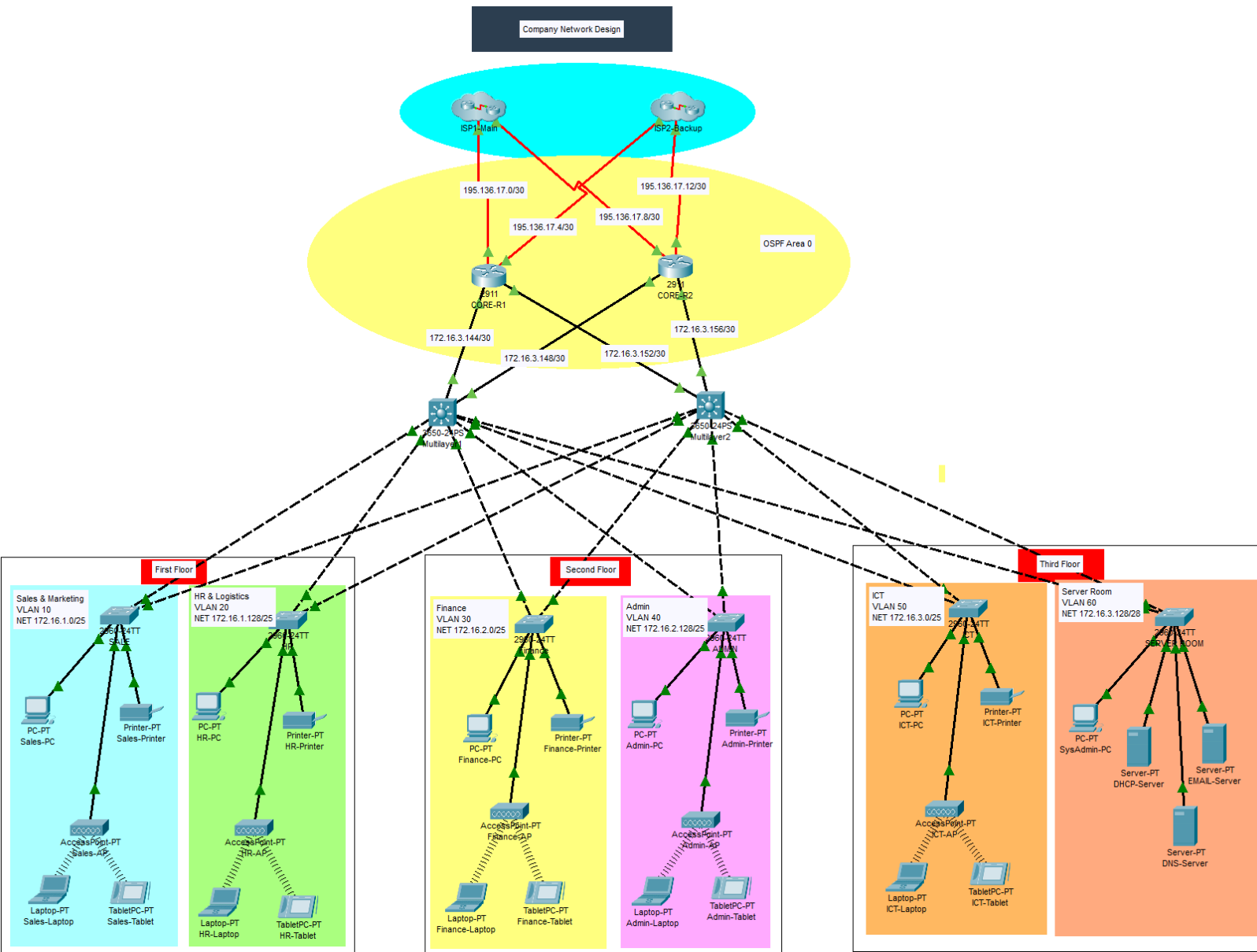
Technologies Used

- **Cisco Packet Tracer:** For designing and simulating the network topology.
- **Routers:** Cisco routers (R1 and R2) for core layer redundancy and ISP connectivity.
- **Multilayer Switches:** Cisco multilayer switches (MLS-W1 and MLS-W2) for distribution layer redundancy and inter-VLAN routing.

- **Access Switches:** Cisco access switches for connecting end-user devices.
- **Wireless Access Points:** For providing wireless networking in each department.
- **Protocols:**
 - **OSPF:** For routing between routers and switches.
 - **SSH:** For secure remote access to routers and switches.
 - **PAT:** For internet access translation.
 - **DHCP:** For dynamic IP address allocation.
 - **VLANs:** For network segmentation.

Conclusion

The designed network architecture provides a robust, scalable, and secure solution for the organization's expansion to a new building. The hierarchical model with redundancy at every layer ensures high availability and performance, while VLAN segmentation and subnetting provide efficient network management. The use of OSPF, SSH, and PAT ensures efficient routing, secure remote access, and controlled internet access. The network design is future-proofed to accommodate potential growth and changes, making it a reliable foundation for the organization's continued success.



IP Addressing

Base Network: 172.16.1.0

First Floor

Department	Network Address	Subnet Mask	Host Address Range	Broadcast Address
Sales & Marketing	172.16.1.0	255.255.255.128/25	172.16.1.1 to 172.16.1.126	172.16.1.127
HR and Logistics	172.16.1.128	255.255.255.128/25	172.16.1.129 to 172.16.1.254	172.16.1.255

Second Floor

Department	Network Address	Subnet Mask	Host Address Range	Broadcast Address
Finance & Accounts	172.16.2.0	255.255.255.128/25	172.16.2.1 to 172.16.2.126	172.16.2.127
Admin & Public Relations	172.16.2.128	255.255.255.128/25	172.16.2.129 to 172.16.2.254	172.16.2.255

Third Floor

Department	Network Address	Subnet Mask	Host Address Range	Broadcast Address
ICT	172.16.3.0	255.255.255.128/25	172.16.3.1 to 172.16.3.126	172.16.3.127
Server Room	172.16.3.128	255.255.255.240/25	172.16.3.129 to 172.16.1.142	172.16.3.143

Between the Routers and Layer-3 Switches

No.	Network Address	Subnet Mask	Host Address Range	Broadcast Address
R1-MLSW1	172.16.3.144	255.255.255.252	172.16.3.145 to 172.16.3.147	172.16.3.147
R1-MLSW2	172.16.3.148	255.255.255.252	172.16.3.149 to 172.16.3.151	172.16.3.151
R2-MLSW1	172.16.3.152	255.255.255.252	172.16.3.153 to 172.16.3.155	172.16.3.155
R2-MLSW2	172.16.3.156	255.255.255.252	172.16.3.157 to 172.16.3.158	172.16.3.159

Between the Routers and ISPs

Public IP addresses 195.136.17.0/30, 195.136.17.4/30, 195.136.17.8/30 and 195.136.17.12/30

Sales Switch

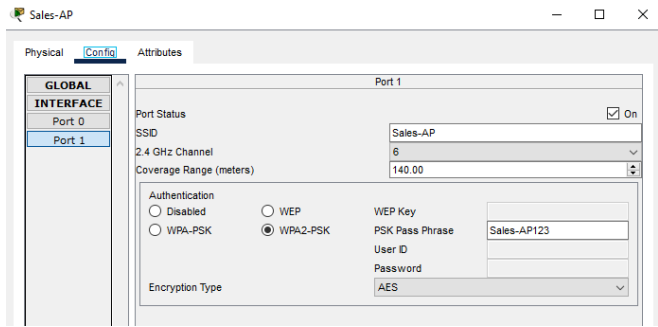
```
Switch>enable
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname Sales-SW
Sales-SW(config)#banner motd #NO Unauthorized Access!!!#
Sales-SW(config)#no ip domain lookup
Sales-SW(config)#line console 0
Sales-SW(config-line)#passw cisco
Sales-SW(config-line)#login
Sales-SW(config-line)#exit
Sales-SW(config)#exit
Sales-SW#
%SYS-5-CONFIG_I: Configured from console by console

Sales-SW#config t
Enter configuration commands, one per line. End with CNTL/Z.
Sales-SW(config)#enable pass
% Incomplete command.
Sales-SW(config)#enable password cisco
Sales-SW(config)#service pas
Sales-SW(config)#service password-encryption
Sales-SW(config)#ex
Sales-SW#
%SYS-5-CONFIG_I: Configured from console by console

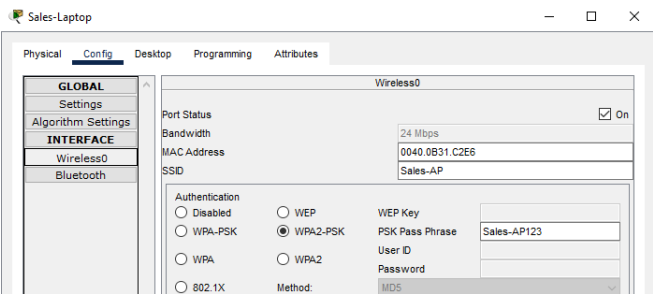
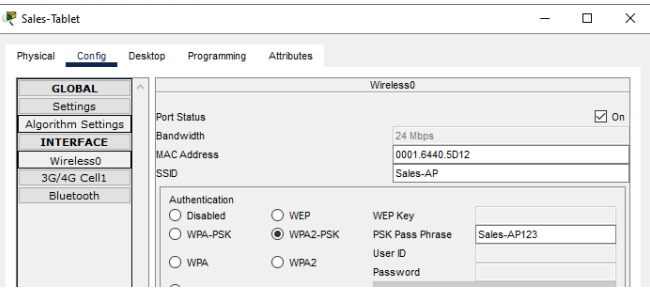
Sales-SW#vr
Building configuration...
[OK]
Sales-SW#
Sales-SW(config)#int range fa0/1-2
Sales-SW(config-if-range)#switchport mode trunk

Sales-SW(config-if-range)#exit
Sales-SW(config)#
Sales-SW(config)#vlan 10
Sales-SW(config-vlan)#name Sales
Sales-SW(config-vlan)#vlan 99
Sales-SW(config-vlan)#name BlackHole
Sales-SW(config-vlan)#exit
Sales-SW(config)#
Sales-SW(config)#int range fa0/3-24
Sales-SW(config-if-range)#switchport mode access
Sales-SW(config-if-range)#switchport access vlan 10
Sales-SW(config-if-range)#exit
Sales-SW(config)#
Sales-SW(config)#int range gig0/1-2
Sales-SW(config-if-range)#switchport mode access
Sales-SW(config-if-range)#switchport access vlan 99
Sales-SW(config-if-range)#exit
Sales-SW(config)#
Sales-SW(config)#do wr
Sales-SW#
```

Sales Access Point



Sales Tablet & Laptop



HR switch

```
Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#hostname HR-SW
HR-SW(config)#line console 0
HR-SW(config-line)#password cisco
HR-SW(config-line)#login
HR-SW(config-line)#exit
HR-SW(config)#
HR-SW(config)#enable password cisco
HR-SW(config)#no ip domain-lookup
HR-SW(config)#banner motd #NO Unauthorised Access!!!#
HR-SW(config)#service password-encryption
HR-SW(config)#
HR-SW(config)#do wr
Building configuration...
[OK]
```

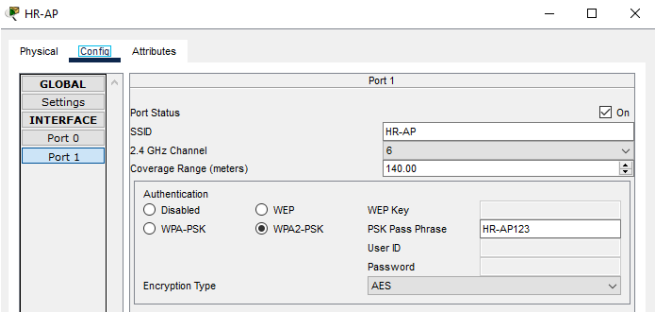
```

HR-SW(config)#int range fa0/1-2
HR-SW(config-if-range)#switchport mode trunk

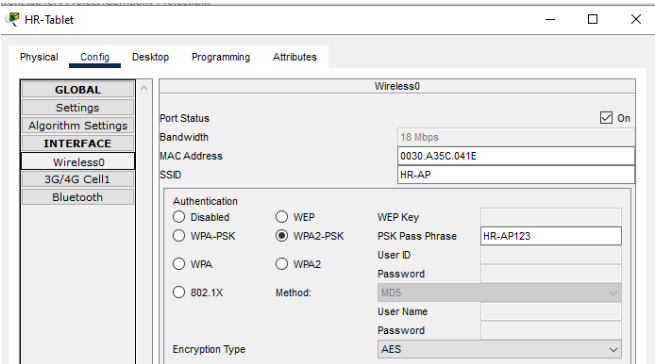
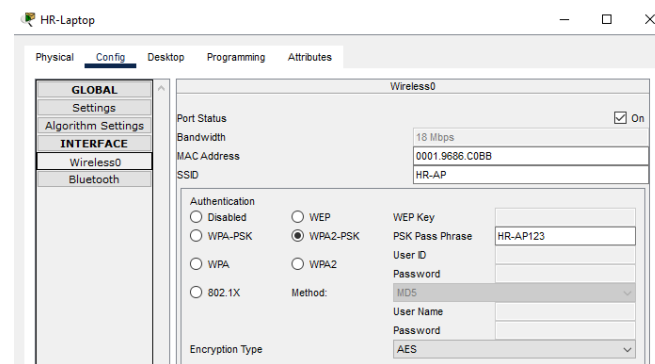
HR-SW(config-if-range)#exit
HR-SW(config)#
HR-SW(config)#vlan 20
HR-SW(config-vlan)#name HR
HR-SW(config-vlan)#vlan 99
HR-SW(config-vlan)#name BlackHole
HR-SW(config-vlan)#exit
HR-SW(config)#
HR-SW(config)#int range fa0/3-24
HR-SW(config-if-range)#switchport mode access
HR-SW(config-if-range)#switchport access vlan 20
HR-SW(config-if-range)#exit
HR-SW(config)#
HR-SW(config)#int range gig0/1-2
HR-SW(config-if-range)#switchport mode access
HR-SW(config-if-range)#switchport access vlan 99
HR-SW(config-if-range)#exit
HR-SW(config)#
HR-SW(config)#do wr

```

HR access point



HR Tablet & Laptop



Finance Switch

```

Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#hostname Finance-SW
Finance-SW(config)#line console 0
Finance-SW(config-line)#password cisco
Finance-SW(config-line)#login
Finance-SW(config-line)#exit
Finance-SW(config)#
Finance-SW(config)#enable password cisco
Finance-SW(config)#no ip domain-lookup
Finance-SW(config)#banner motd #NO Unauthorised Access!!!#
Finance-SW(config)#service password-encryption
Finance-SW(config)#int range fa0/1-2
Finance-SW(config-if-range)#switchport mode trunk

Finance-SW(config-if-range)#exit
Finance-SW(config)#
Finance-SW(config)#vlan 30
Finance-SW(config-vlan)#name Finance
Finance-SW(config-vlan)#vlan 99
Finance-SW(config-vlan)#name BlackHole
Finance-SW(config-vlan)#exit
Finance-SW(config)#
Finance-SW(config)#int range fa0/3-24
Finance-SW(config-if-range)#switchport mode access
Finance-SW(config-if-range)#switchport access vlan 30
Finance-SW(config-if-range)#exit
Finance-SW(config)#
Finance-SW(config)#int range gig0/1-2
Finance-SW(config-if-range)#switchport mode access
Finance-SW(config-if-range)#switchport access vlan 99
Finance-SW(config-if-range)#exit
Finance-SW(config)#
Finance-SW(config)#do wr
Building configuration...
[OK]
Finance-SW(config)#
Finance-SW(config)#do sh port-security

```

```

Finance-SW#config t
Enter configuration commands, one per line. End with CNTL/Z.
Finance-SW(config)#interface range fa0/3-24
Finance-SW(config-if-range)#switchport port-security
Finance-SW(config-if-range)#exit
Finance-SW(config)#do sh port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
(Count) (Count) (Count)
-----
Fa0/3 1 0 0 Shutdown
Fa0/4 1 0 0 Shutdown
Fa0/5 1 0 0 Shutdown
Fa0/6 1 0 0 Shutdown
Fa0/7 1 0 0 Shutdown
Fa0/8 1 0 0 Shutdown
Fa0/9 1 0 0 Shutdown
Fa0/10 1 0 0 Shutdown
Fa0/11 1 0 0 Shutdown
Fa0/12 1 0 0 Shutdown
Fa0/13 1 0 0 Shutdown
Fa0/14 1 0 0 Shutdown
Fa0/15 1 0 0 Shutdown
Fa0/16 1 0 0 Shutdown
Fa0/17 1 0 0 Shutdown
Fa0/18 1 0 0 Shutdown
Fa0/19 1 0 0 Shutdown
Fa0/20 1 0 0 Shutdown
Fa0/21 1 0 0 Shutdown
Fa0/22 1 0 0 Shutdown
Fa0/23 1 0 0 Shutdown
Fa0/24 1 0 0 Shutdown

```

```

Finance-SW#config t
Enter configuration commands, one per line. End with CNTL/Z.
Finance-SW(config)#interface FastEthernet0/5
Finance-SW(config-if)#no switchport port-security
Finance-SW(config-if)#no switchport port-security maximum 1
Finance-SW(config-if)#no switchport port-security mac-address sticky
Finance-SW(config-if)#no switchport port-security violation shutdown
Finance-SW(config-if)#do wr
Building configuration...
[OK]

```

Finance access point

The screenshot shows the 'Finance-AP' configuration window with the 'Config' tab selected. Under the 'INTERFACE' section, 'Port 1' is configured. The 'Port Status' is 'On'. The 'SSID' is 'Finance-AP', the '2.4 GHz Channel' is '6', and the 'Coverage Range (meters)' is '140.00'. Under 'Authentication', 'WPA2-PSK' is selected. The 'WEP Key' is empty, 'PSK Pass Phrase' is 'Finance-API23', 'User ID' is empty, and 'Password' is empty. The 'Encryption Type' is 'AES'.

Finance Tablet & Laptop

The screenshot shows the 'Finance-Tablet' configuration window with the 'Config' tab selected. Under the 'INTERFACE' section, 'Wireless0' is configured. The 'Port Status' is 'On', 'Bandwidth' is '24 Mbps', 'MAC Address' is '0003.E423.1B03', and 'SSID' is 'Finance-AP'. Under 'Authentication', 'WPA2-PSK' is selected. The 'WEP Key' is empty, 'PSK Pass Phrase' is 'Finance-API23', 'User ID' is empty, 'Password' is empty, and 'Method' is 'MD5'. The 'Encryption Type' is 'AES'.

The screenshot shows the 'Finance-Laptop' configuration window with the 'Config' tab selected. Under the 'INTERFACE' section, 'Wireless0' is configured. The 'Port Status' is 'On', 'Bandwidth' is '18 Mbps', 'MAC Address' is '000B.BECC.BEED', and 'SSID' is 'Finance-AP'. Under 'Authentication', 'WPA2-PSK' is selected. The 'WEP Key' is empty, 'PSK Pass Phrase' is 'Finance-API23', 'User ID' is empty, 'Password' is empty, and 'Method' is 'MD5'. The 'Encryption Type' is 'AES'.

Admin

```

Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#hostname Admin-SW
Admin-SW(config)#line console 0
Admin-SW(config-line)#password cisco
Admin-SW(config-line)#login
Admin-SW(config-line)#exit
Admin-SW(config)#
Admin-SW(config)#enable password cisco
Admin-SW(config)#no ip domain-lookup
Admin-SW(config)#banner motd #NO Unauthorized Access!!!#
Admin-SW(config)#service password-encryption
Admin-SW(config)#int range fa0/1-2
Admin-SW(config-if-range)#switchport mode trunk

```

```

Admin-SW(config-if-range)#exit
Admin-SW(config)#
Admin-SW(config)#vlan 40
Admin-SW(config-vlan)#name Admin
Admin-SW(config-vlan)#vlan 99
Admin-SW(config-vlan)#name BlackHole
Admin-SW(config-vlan)#exit
Admin-SW(config)#
Admin-SW(config)#int range fa0/3-24
Admin-SW(config-if-range)#switchport mode access
Admin-SW(config-if-range)#switchport access vlan 40
Admin-SW(config-if-range)#exit
Admin-SW(config)#
Admin-SW(config)#int range gig0/1-2
Admin-SW(config-if-range)#switchport mode access
Admin-SW(config-if-range)#switchport access vlan 99
Admin-SW(config-if-range)#exit
Admin-SW(config)#
Admin-SW(config)#do wr

```

Admin access point

Admin-AP

PhysicalConfigAttributes

GLOBAL

Settings

INTERFACE

Port 0

Port 1

Port 1

Port Status

SSID

2.4 GHz Channel

Coverage Range (meters)

Authentication

Encryption Type

On

Admin-AP

6

140.00

Disabled

WEP

WPA-PSK

WPA2-PSK

WEP Key

PSK Pass Phrase

User ID

Password

Admin-AP123

AES

Admin Tablet & Laptop

Admin-Laptop

PhysicalConfigDesktopProgrammingAttributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

Wireless0

Bluetooth

Wireless0

Port Status

Bandwidth

MAC Address

SSID

Authentication

Encryption Type

IP Configuration

On

16 Mbps

00D0.FFAE.1994

Admin-AP

Disabled

WPA-PSK

WPA

802.1X

WEP

WPA2-PSK

WPA2

Method:

WEP Key

PSK Pass Phrase

User ID

Password

MD5

User Name

Password

Admin-AP123

AES

DHCP

Static

IPv4 Address

Subnet Mask

172.16.2.134

255.255.255.128

Admin-Tablet

PhysicalConfigDesktopProgrammingAttributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

Wireless0

3G/4G Cell1

Bluetooth

Wireless0

Port Status

Bandwidth

MAC Address

SSID

Authentication

Encryption Type

IP Configuration

On

12 Mbps

0010.11D1.E00A

Admin-AP

Disabled

WPA-PSK

WPA

802.1X

WEP

WPA2-PSK

WPA2

Method:

WEP Key

PSK Pass Phrase

User ID

Password

MD5

User Name

Password

Admin-AP123

AES

DHCP

Static

IPv4 Address

Subnet Mask

172.16.2.137

255.255.255.128

ICT Switch

```
Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#hostname ICT-SW
ICT-SW(config)#line console 0
ICT-SW(config-line)#password cisco
ICT-SW(config-line)#login
ICT-SW(config-line)#exit
ICT-SW(config)#
ICT-SW(config)#enable password cisco
ICT-SW(config)#no ip domain-lookup
ICT-SW(config)#banner motd #NO Unauthorised Access!!!#
ICT-SW(config)#service password-encryption
ICT-SW(config)#int range fa0/1-2
ICT-SW(config-if-range)#switchport mode trunk

ICT-SW(config-if-range)#exit
ICT-SW(config)#
ICT-SW(config)#vlan 50
ICT-SW(config-vlan)#name ICT
ICT-SW(config-vlan)#vlan 99
ICT-SW(config-vlan)#name BlackHole
ICT-SW(config-vlan)#exit
ICT-SW(config)#
ICT-SW(config)#int range fa0/3-24
ICT-SW(config-if-range)#switchport mode access
ICT-SW(config-if-range)#switchport access vlan 50
ICT-SW(config-if-range)#exit
ICT-SW(config)#
ICT-SW(config)#int range gig0/1-2
ICT-SW(config-if-range)#switchport mode access
ICT-SW(config-if-range)#switchport access vlan 99
ICT-SW(config-if-range)#exit
ICT-SW(config)#
ICT-SW(config)#do wr
-----
```

ICT Access Point

ICT-AP

PhysicalConfigAttributes

GLOBAL

Settings

INTERFACE

Port 0

Port 1

Port 1

Port Status

SSID

2.4 GHz Channel

Coverage Range (meters)

Authentication

Encryption Type

On

ICT-AP

6

140.00

Disabled

WEP

WPA-PSK

WPA2-PSK

WEP Key

PSK Pass Phrase

User ID

Password

ICT-AP123

AES

ICT Tablet & Laptop

Physical Config Desktop Programming Attributes

GLOBAL Settings Algorithm Settings INTERFACE

Wireless0

Bluetooth

Port Status ☒ On

Bandwidth 6 Mbps

MAC Address 000D.B08C.A642

SSID ICT-AP

Authentication ☐ Disabled ☐ WEP ☐ WPA-PSK ☒ WPA2-PSK ☐ WPA ☐ WPA2 ☐ 802.1X

WEP Key

PSK Pass Phrase ICT-AP123

User ID

Password

User Name

Method: MD5

Encryption Type AES

Physical Config Desktop Programming Attributes

GLOBAL Settings Algorithm Settings INTERFACE

Wireless0

Bluetooth

Port Status ☒ On

Bandwidth 1 Mbps

MAC Address 0090.2190.CC79

SSID ICT-AP

Authentication ☐ Disabled ☐ WEP ☐ WPA-PSK ☒ WPA2-PSK ☐ WPA ☐ WPA2 ☐ 802.1X

WEP Key

PSK Pass Phrase ICT-AP123

User ID

Password

User Name

Method: MD5

Encryption Type AES

Server Room Switch

```
Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#hostname Server-Room-SW
Server-Room-SW(config)#line console 0
Server-Room-SW(config-line)#password cisco
Server-Room-SW(config-line)#login
Server-Room-SW(config-line)#exit
Server-Room-SW(config)#
Server-Room-SW(config)#enable password cisco
Server-Room-SW(config)#no ip domain-lookup
Server-Room-SW(config)#banner motd #NO Unauthorised Access!!!#
Server-Room-SW(config)#service password-encryption
Server-Room-SW(config)#
Server-Room-SW(config)#int range fa0/1-2
Server-Room-SW(config-if-range)#switchport mode trunk
```

```
Server-Room-SW(config-if-range)#exit
Server-Room-SW(config)#
Server-Room-SW(config)#vlan 60
Server-Room-SW(config-vlan)#name ServerRoom
Server-Room-SW(config-vlan)#vlan 99
Server-Room-SW(config-vlan)#name BlackHole
Server-Room-SW(config-vlan)#exit
Server-Room-SW(config)#
Server-Room-SW(config)#int range fa0/3-24
Server-Room-SW(config-if-range)#switchport mode access
Server-Room-SW(config-if-range)#switchport access vlan 60
Server-Room-SW(config-if-range)#exit
Server-Room-SW(config)#
Server-Room-SW(config)#int range gig0/1-2
Server-Room-SW(config-if-range)#switchport mode access
Server-Room-SW(config-if-range)#switchport access vlan 99
Server-Room-SW(config-if-range)#exit
Server-Room-SW(config)#
Server-Room-SW(config)#do wr
```

DHCP Server

Physical Config Services Desktop Programming Attributes

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 172.16.3.130

Subnet Mask 255.255.255.240

Default Gateway 172.16.3.129

DNS Server 172.16.3.131

IPv6 Configuration

Physical Config Services Desktop Programming Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DHCP

Interface FastEthernet0 Service ☐ On ☒ Off

Pool Name SalesPool

Default Gateway 172.16.1.1

DNS Server 172.16.3.131

Start IP Address: 172.16.1.1

Subnet Mask: 255.255.255.128

Maximum Number of Users: 120

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
SalesPool	172.16.1.1	172.16.3.131	172.16.1.1	255.255.255.128	120	0.0.0.0	0.0.0.0

Physical Config Services Desktop Programming Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DHCP

Interface FastEthernet0 Service ☐ On ☒ Off

Pool Name HRPool

Default Gateway 172.16.1.129

DNS Server 172.16.3.131

Start IP Address: 172.16.1.1

Subnet Mask: 255.255.255.128

Maximum Number of Users: 120

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
HRPool	172.16.1.129	172.16.3.131	172.16.1.1	255.255.255.128	120	0.0.0.0	0.0.0.0

Physical Config Services Desktop Programming Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DHCP

Interface FastEthernet0 Service ☐ On ☒ Off

Pool Name FinancePool

Default Gateway 172.16.2.1

DNS Server 172.16.3.131

Start IP Address: 172.16.2.1

Subnet Mask: 255.255.255.128

Maximum Number of Users: 120

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
FinancePool	172.16.2.1	172.16.3.131	172.16.2.1	255.255.255.128	120	0.0.0.0	0.0.0.0

DHCP-Server

PhysicalConfigServicesDesktopProgrammingAttributes

SERVICES

HTTPDHCPDHCPv6TFTPDNSSYSLOGAAANTPEMAILFTPVM ManagementRadius EAP

DHCP

InterfaceFastEthernet0ServiceOnOff

Pool NameAdminPool

Default Gateway172.16.2.129

DNS Server172.16.3.131

Start IP Address:172162134

Subnet Mask:255255255128

Maximum Number of Users:120

TFTP Server:0.0.0.0

WLC Address:0.0.0.0

AddSaveRemove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
FinancePool	172.16.2.0	172.16.3...	172.16.2.6	255.255...	120	0.0.0.0	0.0.0.0
AdminPool	172.16.2...	172.16.3...	172.16.2...	255.255...	120	0.0.0.0	0.0.0.0

DHCP-Server

PhysicalConfigServicesDesktopProgrammingAttributes

SERVICES

HTTPDHCPDHCPv6TFTPDNSSYSLOGAAANTPEMAILFTPVM ManagementRadius EAP

DHCP

InterfaceFastEthernet0ServiceOnOff

Pool NameICTPool

Default Gateway172.16.3.1

DNS Server172.16.3.131

Start IP Address:1721636

Subnet Mask:255255255128

Maximum Number of Users:120

TFTP Server:0.0.0.0

WLC Address:0.0.0.0

AddSaveRemove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
ICTPool	172.16.3.1	172.16.3...	172.16.3.6	255.255...	120	0.0.0.0	0.0.0.0

DNS-Server

DNS-Server

PhysicalConfigServicesDesktopProgrammingAttributes

IP Configuration

IP Configuration

Static

IPv4 Address172.16.3.131

Subnet Mask255.255.255.240

Default Gateway172.16.3.129

DNS Server0.0.0.0

IPv6 Configuration

DNS-Server

PhysicalConfigServicesDesktopProgrammingAttributes

SERVICES

HTTPDHCPDHCPv6TFTPDNSSYSLOGAAANTPEMAILFTPVM ManagementRadius EAP

DNS

DNS ServiceOnOff

Resource Records

NameTypeA Record

Address

AddSaveRemove

No.	Name	Type	Detail
0	www.gtech.com	A Record	172.16.3.131

MultiLayer Switch-1

```
Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#hostname Mlt-SW1
Mlt-SW1(config)#line console 0
Mlt-SW1(config-line)#password cisco
Mlt-SW1(config-line)#login
Mlt-SW1(config-line)#exit
Mlt-SW1(config)#
Mlt-SW1(config)#enable password cisco
Mlt-SW1(config)#no ip domain-lookup
Mlt-SW1(config)#banner motd #NO Unauthorised Access!!!#
Mlt-SW1(config)#service password-encryption
Mlt-SW1(config)#
Mlt-SW1(config)#ip domain name
% Incomplete command.
Mlt-SW1(config)#ip domain name cisco.net
Mlt-SW1(config)#user admin password cisco
Mlt-SW1(config)#crypto key generate rsa
The name for the keys will be: Mlt-SW1.cisco.net
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

Mlt-SW1(config)#crypto key generate rsa
*Mar 1 3:24:29.478: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 3:24:29.478: %SSH-5-ENABLED: SSH 1.5 has been enabled
% You already have RSA keys defined named Mlt-SW1.cisco.net .
% Do you really want to replace them? [yes/no]: yes
The name for the keys will be: Mlt-SW1.cisco.net
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

Mlt-SW1(config)#line vty 0 15
*Mar 1 3:25:1.512: %SSH-5-ENABLED: SSH 1.99 has been enabled
Mlt-SW1(config-line)#login local
Mlt-SW1(config-line)#transport input ssh
Mlt-SW1(config-line)#ex
% Ambiguous command: "ex"
Mlt-SW1(config-line)#exit

Mlt-SW1(config)#ip ssh version 2
...
Mlt-SW1(config)#int range gig1/0/3-8
Mlt-SW1(config-if-range)#switchport mode trunk
Mlt-SW1(config-if-range)#
Mlt-SW1(config-if-range)#vlan 10
Mlt-SW1(config-vlan)#name Sales
Mlt-SW1(config-vlan)#vlan 20
Mlt-SW1(config-vlan)#name HR
Mlt-SW1(config-vlan)#vlan 30
Mlt-SW1(config-vlan)#name Finance
Mlt-SW1(config-vlan)#vlan 40
Mlt-SW1(config-vlan)#name Admin
Mlt-SW1(config-vlan)#vlan 50
Mlt-SW1(config-vlan)#name ICT
Mlt-SW1(config-vlan)#vlan 60
Mlt-SW1(config-vlan)#name ServerRoom
Mlt-SW1(config-vlan)#
Mlt-SW1(config-vlan)#exit
Mlt-SW1(config)#
Mlt-SW1(config)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
```

```

Mlt-SW1(config)#int range gig1/0/1-2
Mlt-SW1(config-if-range)#no switchport
Mlt-SW1(config-if-range)#ex
Mlt-SW1(config)#
Mlt-SW1(config)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
Mlt-SW1(config)#
Mlt-SW1(config)#ex
Mlt-SW1#
Mlt-SW1(config)#int gig1/0/1
Mlt-SW1(config-if)#ip address 172.16.3.145 255.255.255.252
Mlt-SW1(config-if)#no sh
Mlt-SW1(config-if)#
Mlt-SW1(config-if)#ex
Mlt-SW1(config)#
Mlt-SW1(config)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
Mlt-SW1(config)#
Mlt-SW1(config)#int gig1/0/2
Mlt-SW1(config-if)#ip address 172.16.3.149 255.255.255.252
Mlt-SW1(config-if)#no sh
Mlt-SW1(config-if)#
Mlt-SW1(config-if)#
Mlt-SW1(config-if)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
Mlt-SW1(config-if)#
Mlt-SW1(config)#ip routing
Mlt-SW1(config)#router ospf 10
Mlt-SW1(config-router)#router-id 2.2.2.2
Mlt-SW1(config-router)#network 172.16.1.0 0.0.0.127 area 0
Mlt-SW1(config-router)#network 172.16.1.128 0.0.0.127 area 0
Mlt-SW1(config-router)#network 172.16.2.0 0.0.0.127 area 0
Mlt-SW1(config-router)#network 172.16.2.128 0.0.0.127 area 0
Mlt-SW1(config-router)#network 172.16.3.0 0.0.0.127 area 0
Mlt-SW1(config-router)#network 172.16.3.128 0.0.0.15 area 0
Mlt-SW1(config-router)#
Mlt-SW1(config-router)#network 172.16.3.144 0.0.0.3 area 0
Mlt-SW1(config-router)#network 172.16.3.148 0.0.0.3 area 0
Mlt-SW1(config-router)#exit
Mlt-SW1(config)#
Mlt-SW1(config)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
Mlt-SW1(config)#int vlan 10
Mlt-SW1(config-if)#no sh
Mlt-SW1(config-if)#ip add 172.16.1.1 255.255.255.128
Mlt-SW1(config-if)#ip helper-address 172.16.3.130
Mlt-SW1(config-if)#ex
Mlt-SW1(config)#
Mlt-SW1(config)#int vlan 20
Mlt-SW1(config-if)#no sh
Mlt-SW1(config-if)#ip add 172.16.1.129 255.255.255.128
Mlt-SW1(config-if)#ip helper-address 172.16.3.130
Mlt-SW1(config-if)#ex
Mlt-SW1(config)#
Mlt-SW1(config)#int vlan 30
Mlt-SW1(config-if)#no sh
Mlt-SW1(config-if)#ip add 172.16.2.1 255.255.255.128
Mlt-SW1(config-if)#ip helper-address 172.16.3.130
Mlt-SW1(config-if)#ex
Mlt-SW1(config)#
Mlt-SW1(config)#int vlan 40
Mlt-SW1(config-if)#no sh
Mlt-SW1(config-if)#ip add 172.16.2.129 255.255.255.128
Mlt-SW1(config-if)#ip helper-address 172.16.3.130
Mlt-SW1(config-if)#ex
Mlt-SW1(config)#
Mlt-SW1(config)#int vlan 50
Mlt-SW1(config-if)#no sh
Mlt-SW1(config-if)#ip add 172.16.3.1 255.255.255.128
Mlt-SW1(config-if)#ip helper-address 172.16.3.130
Mlt-SW1(config-if)#ex
Mlt-SW1(config)#
Mlt-SW1(config)#int vlan 60
Mlt-SW1(config-if)#no sh
Mlt-SW1(config-if)#ip add 172.16.3.129 255.255.255.240
Mlt-SW1(config-if)#ip helper-address 172.16.3.130
Mlt-SW1(config-if)#ex
Mlt-SW1(config)#ip route 0.0.0.0 0.0.0.0 gig1/0/1
Mlt-SW1(config)#ip route 0.0.0.0 0.0.0.0 gig1/0/2 70
Mlt-SW1(config)#
Mlt-SW1(config)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]

```

MultiLayer Switch-2

```

Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#hostname Mlt-SW2
Mlt-SW2(config)#line console 0
Mlt-SW2(config-line)#password cisco
Mlt-SW2(config-line)#login
Mlt-SW2(config-line)#exit
Mlt-SW2(config)#
Mlt-SW2(config)#enable password cisco
Mlt-SW2(config)#no ip domain-lookup
Mlt-SW2(config)#banner motd #NO Unauthorised Access!!!#
Mlt-SW2(config)#service password-encryption
Mlt-SW2(config)#
Mlt-SW2(config)#do wr
Building configuration...
[OK]

```

```

Mlt-SW2(config)#
Mlt-SW2(config)#ip domain name cisco.net
Mlt-SW2(config)#username admin password cisco
Mlt-SW2(config)#crypto key generate rsa
The name for the keys will be: Mlt-SW2.cisco.net
Choose the size of the key modulus in the range of 360 to 2048 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

Mlt-SW2(config)#line vty 0 15
*Mar 1 3:32:18.512: %SSH-5-ENABLED: SSH 1.99 has been enabled
Mlt-SW2(config-line)#login local
Mlt-SW2(config-line)#transport input ssh
Mlt-SW2(config-line)#exit
Mlt-SW2(config)#
Mlt-SW2(config)#do wr
Building configuration...
[OK]

Mlt-SW2(config)#ip ssh version 2
Mlt-SW2(config)#do wr
Building configuration...
[OK]

Mlt-SW2(config)#int range gig1/0/3-8
Mlt-SW2(config-if-range)#switchport mode trunk
Mlt-SW2(config-if-range)#
Mlt-SW2(config-if-range)#vlan 10
Mlt-SW2(config-vlan)#name Sales
Mlt-SW2(config-vlan)#vlan 20
Mlt-SW2(config-vlan)#name HR
Mlt-SW2(config-vlan)#vlan 30
Mlt-SW2(config-vlan)#name Finance
Mlt-SW2(config-vlan)#vlan 40
Mlt-SW2(config-vlan)#name Admin
Mlt-SW2(config-vlan)#vlan 50
Mlt-SW2(config-vlan)#name ICT
Mlt-SW2(config-vlan)#vlan 60
Mlt-SW2(config-vlan)#name ServerRoom
Mlt-SW2(config-vlan)#
Mlt-SW2(config-vlan)#exit
Mlt-SW2(config)#
Mlt-SW2(config)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]

Mlt-SW2(config)#int range gig1/0/1-2
Mlt-SW2(config-if-range)#no switchport
Mlt-SW2(config-if-range)#ex
Mlt-SW2(config)#
Mlt-SW2(config)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]

Mlt-SW2(config)#
Mlt-SW2(config)#ex
Mlt-SW2#

Mlt-SW2(config)#int gig1/0/1
Mlt-SW2(config-if)#ip address 172.16.3.153 255.255.255.252
Mlt-SW2(config-if)#no sh
Mlt-SW2(config-if)#
Mlt-SW2(config-if)#ex
Mlt-SW2(config)#
Mlt-SW2(config)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]

Mlt-SW2(config)#
Mlt-SW2(config)#int gig1/0/2
Mlt-SW2(config-if)#ip address 172.16.3.157 255.255.255.252
Mlt-SW2(config-if)#no sh
Mlt-SW2(config-if)#
Mlt-SW2(config-if)#
Mlt-SW2(config)#ip routing
Mlt-SW2(config)#router ospf 10
Mlt-SW2(config-router)#network 172.16.1.0 0.0.0.127 area 0
Mlt-SW2(config-router)#network 172.16.1.128 0.0.0.127 area 0
Mlt-SW2(config-router)#network 172.16.2.0 0.0.0.127 area 0
Mlt-SW2(config-router)#network 172.16.2.128 0.0.0.127 area 0
Mlt-SW2(config-router)#network 172.16.3.0 0.0.0.127 area 0
Mlt-SW2(config-router)#network 172.16.3.128 0.0.0.15 area 0
Mlt-SW2(config-router)#network 172.16.3.152 0.0.0.3 area 0
Mlt-SW2(config-router)#network 172.16.3.156 0.0.0.3 area 0
Mlt-SW2(config-router)#exit
Mlt-SW2(config)#
Mlt-SW2(config)#router ospf 10
Mlt-SW2(config-router)#router-id 1.1.1.1
Mlt-SW2(config-router)#Reload or use "clear ip ospf process" command, for this to take
effect

Mlt-SW2(config)#int vlan 10
Mlt-SW2(config-if)#no sh
Mlt-SW2(config-if)#ip add 172.16.1.1 255.255.255.128
Mlt-SW2(config-if)#ip helper-address 172.16.3.130
Mlt-SW2(config-if)#ex
Mlt-SW2(config)#
Mlt-SW2(config)#int vlan 20
Mlt-SW2(config-if)#no sh
Mlt-SW2(config-if)#ip add 172.16.1.129 255.255.255.128
Mlt-SW2(config-if)#ip helper-address 172.16.3.130
Mlt-SW2(config-if)#ex
Mlt-SW2(config)#
Mlt-SW2(config)#int vlan 30
Mlt-SW2(config-if)#no sh
Mlt-SW2(config-if)#ip add 172.16.2.1 255.255.255.128
Mlt-SW2(config-if)#ip helper-address 172.16.3.130

Mlt-SW2(config)#int vlan 40
Mlt-SW2(config-if)#no sh
Mlt-SW2(config-if)#ip add 172.16.2.129 255.255.255.128
Mlt-SW2(config-if)#ip helper-address 172.16.3.130
Mlt-SW2(config-if)#ex
Mlt-SW2(config)#
Mlt-SW2(config)#int vlan 50
Mlt-SW2(config-if)#no sh
Mlt-SW2(config-if)#ip add 172.16.3.1 255.255.255.128
Mlt-SW2(config-if)#ip helper-address 172.16.3.130
Mlt-SW2(config-if)#ex
Mlt-SW2(config)#
Mlt-SW2(config)#int vlan 60
Mlt-SW2(config-if)#no sh
Mlt-SW2(config-if)#ip add 172.16.3.129 255.255.255.240
Mlt-SW2(config-if)#ip helper-address 172.16.3.130
Mlt-SW2(config-if)#ex
Mlt-SW2(config)#

```

```

Mlt-SW2(config)#ip route 0.0.0.0 0.0.0.0 gig1/0/1
Mlt-SW2(config)#ip route 0.0.0.0 0.0.0.0 gig1/0/2 70
Mlt-SW2(config)#
Mlt-SW2(config)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]

```

Core router 1

```

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#hostname CORE-R1
CORE-R1(config)#line console 0
CORE-R1(config-line)#password cisco
CORE-R1(config-line)#login
CORE-R1(config-line)#exit
CORE-R1(config)#
CORE-R1(config)#enable password cisco
CORE-R1(config)#no ip domain-lookup
CORE-R1(config)#banner motd #NO Unauthorised Access!!!#
CORE-R1(config)#service password-encryption
CORE-R1(config)#
CORE-R1(config)#ip domain name cisco.net
CORE-R1(config)#username admin password cisco
CORE-R1(config)#crypto key generate rsa
The name for the keys will be: CORE-R1.cisco.net
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

CORE-R1(config)#line vty 0 15
*Mar 1 3:20:32.730: %SSH-5-ENABLED: SSH 1.99 has been enabled
CORE-R1(config-line)#login local
CORE-R1(config-line)#transport input ssh
CORE-R1(config-line)#exit
CORE-R1(config)#ip ssh version 2
CORE-R1(config)#do wr
Building configuration...
[OK]

CORE-R1(config)#interface GigabitEthernet0/0
CORE-R1(config-if)#ip add 172.16.3.146 255.255.255.252
CORE-R1(config-if)#no sh
CORE-R1(config-if)#ex
CORE-R1(config)#
CORE-R1(config)#interface GigabitEthernet0/1
CORE-R1(config-if)#ip add 172.16.3.154 255.255.255.252
CORE-R1(config-if)#no sh
CORE-R1(config-if)#
CORE-R1(config-if)#ex
CORE-R1(config)#do wr
Building configuration...
[OK]
CORE-R1(config)#exit
CORE-R1#

CORE-R1(config)#int se0/2/0
CORE-R1(config-if)#clock rate 64000
CORE-R1(config-if)#ip add 195.136.17.1 255.255.255.252
CORE-R1(config-if)#no sh
CORE-R1(config-if)#ex
CORE-R1(config)#
CORE-R1(config)#int se0/2/1
CORE-R1(config-if)#clock rate 64000
CORE-R1(config-if)#ip add 195.136.17.5 255.255.255.252
CORE-R1(config-if)#no sh
CORE-R1(config-if)#ex
CORE-R1(config)#router ospf 10
CORE-R1(config-router)#router-id 3.3.3.3
CORE-R1(config-router)#
CORE-R1(config-router)#network 172.16.3.144 0.0.0.3 area 0
CORE-R1(config-router)#network 172.16.3.152 0.0.0.3 area 0
CORE-R1(config-router)#
CORE-R1(config-router)#network 195.136.17.0 0.0.0.3 area 0
CORE-R1(config-router)#network 195.136.17.4 0.0.0.3 area 0
CORE-R1(config-router)#
CORE-R1(config-router)#do wr
Building configuration...
[OK]
CORE-R1(config-router)#ex
CORE-R1(config)#
-----
CORE-R1(config)#ip nat inside source list 1 int se0/2/0 overload
CORE-R1(config)#ip nat inside source list 1 int se0/2/1 overload
CORE-R1(config)#
CORE-R1(config)#access-list 1 permit 172.16.1.0 0.0.0.127
CORE-R1(config)#access-list 1 permit 172.16.1.128 0.0.0.127
CORE-R1(config)#
CORE-R1(config)#access-list 1 permit 172.16.2.0 0.0.0.127
CORE-R1(config)#access-list 1 permit 172.16.2.128 0.0.0.127
CORE-R1(config)#
CORE-R1(config)#access-list 1 permit 172.16.3.0 0.0.0.127
CORE-R1(config)#access-list 1 permit 172.16.3.128 0.0.0.15
-----
CORE-R1(config)#int range gig0/0-1
CORE-R1(config-if-range)#ip nat
% Incomplete command.
CORE-R1(config-if-range)#ip nat ?
    inside    Inside interface for address translation
    outside   Outside interface for address translation
CORE-R1(config-if-range)#ip nat
% Incomplete command.
CORE-R1(config-if-range)#
CORE-R1(config-if-range)#ip nat inside ?
    <cr>
CORE-R1(config-if-range)#ip nat inside
CORE-R1(config-if-range)#int range se0/2/0
    ^
% Invalid input detected at '^' marker.

CORE-R1(config-if-range)#ip nat outside

```

```

CORE-R1(config-if-range)#int se0/2/0
CORE-R1(config-if)#ip nat outside
CORE-R1(config-if)#int se0/2/1
CORE-R1(config-if)#ip nat outside
CORE-R1(config-if)#ex
CORE-R1(config)#
CORE-R1(config)#do wr
Building configuration...
[OK]

CORE-R1(config)#ip route 0.0.0.0 0.0.0.0 se0/2/0
%Default route without gateway, if not a point-to-point interface, may impact performance
CORE-R1(config)#ip route 0.0.0.0 0.0.0.0 se0/2/1 70

```

Core router 2

```

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#hostname CORE-R2
CORE-R2(config)#line console 0
CORE-R2(config-line)#password cisco
CORE-R2(config-line)#login
CORE-R2(config-line)#exit
CORE-R2(config)#
CORE-R2(config)#enable password cisco
CORE-R2(config)#no ip domain-lookup
CORE-R2(config)#banner motd #NO Unauthorised Access!!!#
CORE-R2(config)#service password-encryption

CORE-R2(config)#
CORE-R2(config)#ip domain name cisco.net
CORE-R2(config)#username admin password cisco
CORE-R2(config)#crypto key generate rsa
The name for the keys will be: CORE-R2.cisco.net
Choose the size of the key modulus in the range of 360 to 2048 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

CORE-R2(config)#line vty 0 15
*Mar 1 3:22:52.617: %SSH-5-ENABLED: SSH 1.99 has been enabled
CORE-R2(config-line)#login local
CORE-R2(config-line)#transport input ssh
CORE-R2(config-line)#exit

CORE-R2(config)#ip ssh version 2
CORE-R2(config)#do wr
Building configuration...
[OK]

CORE-R2(config)#interface GigabitEthernet0/0
CORE-R2(config-if)#ip add 172.16.3.150 255.255.255.252
CORE-R2(config-if)#no sh
CORE-R2(config-if)#
CORE-R2(config-if)#ex
CORE-R2(config)#
CORE-R2(config)#interface GigabitEthernet0/1
CORE-R2(config-if)#ip add 172.16.3.158 255.255.255.252
CORE-R2(config-if)#no sh
CORE-R2(config-if)#
CORE-R2(config-if)#ex
CORE-R2(config)#
CORE-R2(config)#int se0/2/0
CORE-R2(config-if)#clock rate 64000
CORE-R2(config-if)#ip add 195.136.17.9 255.255.255.252
CORE-R2(config-if)#no sh
CORE-R2(config-if)#ex
CORE-R2(config)#
CORE-R2(config)#int se0/2/1
CORE-R2(config-if)#clock rate 64000
CORE-R2(config-if)#ip add 195.136.17.13 255.255.255.252
CORE-R2(config-if)#no sh
CORE-R2(config-if)#
CORE-R2(config)#router ospf 10
CORE-R2(config-router)#router-id 4.4.4.4
CORE-R2(config-router)#
CORE-R2(config-router)#network 172.16.3.148 0.0.0.3 area 0
CORE-R2(config-router)#network 172.16.3.156 0.0.0.3 area 0
CORE-R2(config-router)#
CORE-R2(config-router)#network 195.136.17.8 0.0.0.3 area 0
CORE-R2(config-router)#network 195.136.17.12 0.0.0.3 area 0
CORE-R2(config-router)#
CORE-R2(config-router)#do wr
Building configuration...
[OK]

CORE-R2(config-router)#ex
CORE-R2(config)#ip nat inside source list 1 int se0/2/0 overload
CORE-R2(config)#ip nat inside source list 1 int se0/2/1 overload
CORE-R2(config)#
CORE-R2(config)#
CORE-R2(config)#access-list 1 permit 172.16.1.0 0.0.0.127
CORE-R2(config)#access-list 1 permit 172.16.1.128 0.0.0.127
CORE-R2(config)#
CORE-R2(config)#access-list 1 permit 172.16.2.0 0.0.0.127
CORE-R2(config)#access-list 1 permit 172.16.2.128 0.0.0.127
CORE-R2(config)#
CORE-R2(config)#access-list 1 permit 172.16.3.0 0.0.0.127
CORE-R2(config)#access-list 1 permit 172.16.3.128 0.0.0.15
CORE-R2(config)#
CORE-R2(config)#int range gig0/0-1
CORE-R2(config-if-range)#ip nat inside
CORE-R2(config-if-range)#
CORE-R2(config-if-range)#
CORE-R2(config-if-range)#exit
CORE-R2(config)#
CORE-R2(config)#int se0/2/0
CORE-R2(config-if)#ip nat outside
CORE-R2(config-if)#int se0/2/1
CORE-R2(config-if)#ip nat outside
CORE-R2(config-if)#ex
CORE-R2(config)#do wr
Building configuration...
[OK]

CORE-R2(config)#ip route 0.0.0.0 0.0.0.0 se0/2/0
%Default route without gateway, if not a point-to-point interface, may impact performance
CORE-R2(config)#ip route 0.0.0.0 0.0.0.0 se0/2/1 70
CORE-R2(config)#
CORE-R2(config)#do wr
Building configuration...
[OK]

```

ISP Router-1

```
Router(config)#interface Serial0/3/0
Router(config-if)#ip address 195.136.17.2 255.255.255.0
Router(config-if)#ip address 195.136.17.2 255.255.255.252
Router(config-if)#ip address 195.136.17.2 255.255.255.252
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/3/1
Router(config-if)#ip address 195.136.17.10 255.255.255.252
Router(config-if)#
Router(config-if)#
Router(config-if)#do wr
Building configuration...
[OK]
Router(config)#router ospf 10
Router(config-router)#router-id 5.5.5.5
Router(config-router)#
Router(config-router)#network 195.136.17.8 0.0.0.3 area 0
Router(config-router)#network 195.136.17.0 0.0.0.3 area 0
Router(config-router)#
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#ex
Router(config)#
```

ISP Router-2

```
Router(config)#interface Serial0/3/0
Router(config-if)#ip address 195.136.17.6 255.255.255.0
Router(config-if)#ip address 195.136.17.6 255.255.255.0
Router(config-if)#ip address 195.136.17.6 255.255.255.252
Router(config-if)#ip address 195.136.17.6 255.255.255.252
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/3/1
Router(config-if)#

Router(config)#interface Serial0/3/1
Router(config-if)#ip address 195.136.17.14 255.255.255.252
Router(config-if)#ip address 195.136.17.14 255.255.255.252
Router(config-if)#do wr
Building configuration...
[OK]
Router(config)#router ospf 10
Router(config-router)#router-id 6.6.6.6
Router(config-router)#
Router(config-router)#network 195.136.17.4 0.0.0.3 area 0
Router(config-router)#network 195.136.17.12 0.0.0.3 area 0
Router(config-router)#
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#ex
Router(config)#
```

Ping Tests

```
C:\>ping 172.16.3.11

Pinging 172.16.3.11 with 32 bytes of data:

Reply from 172.16.3.11: bytes=32 time=1ms TTL=127
Reply from 172.16.3.11: bytes=32 time=1ms TTL=127
Reply from 172.16.3.11: bytes=32 time<1ms TTL=127
Reply from 172.16.3.11: bytes=32 time=1ms TTL=127

Ping statistics for 172.16.3.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 195.136.17.6

Pinging 195.136.17.6 with 32 bytes of data:

Reply from 195.136.17.6: bytes=32 time=1ms TTL=253
Reply from 195.136.17.6: bytes=32 time=1ms TTL=253
Request timed out.
Reply from 195.136.17.6: bytes=32 time=1ms TTL=253

Ping statistics for 195.136.17.6:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>
```

SSH Access

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l admin 172.16.3.154

Password:
% Login invalid

Password:

NO Unauthorised Access!!!

CORE-R1>en
Password:
CORE-R1#
CORE-R1#
CORE-R1#
CORE-R1#
```